



SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - FEB / MARCH 2025

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Artificial Intelligence	Max. Marks	: 100
Course Code	: IS52 (2021,2022 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- For each of the following activities, give a PEAS description of the task environment: i) Playing soccer ii) Playing a tennis match. CO1 (06)
 - Explain the following two categories of Artificial Intelligence definition with respect to thought processes and reasoning: CO1 (06)
 - Acting Rationally
 - Thinking Humanly.
 - With a suitable diagram and example, describe the agent's interaction with environment. CO1 (08)
- Illustrate with a schematic diagram the working principle of simple reflex agent considering two-state vacuum cleaner environment. CO1 (06)
 - Explain how A* Search algorithm minimizes the total estimated solution cost for the following example. CO1 (08)

1	2	3
	4	6
7	5	8

Start state

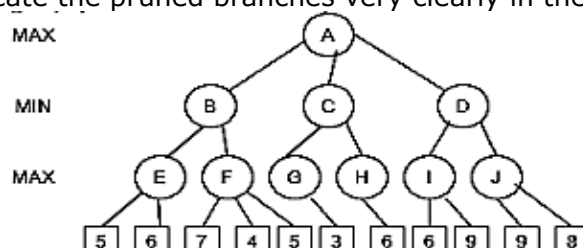
1	2	3
4	5	6
7	8	

Goal State

- Explain the following properties of task environment with example. CO1 (06)
 - Deterministic vs Stochastic
 - Single agent vs Multiagent

UNIT - II

- Define wumpus world problem. Write and interpret the propositional logic to represent the knowledge derived by each of the following sentences for Wumpus world problem: CO2 (10)
 - Square (1, 2) is breezy.
 - No stench in square (2, 1).
 - Glitter and stench in square (3, 3).
 - Explain Syntax and Semantics of Propositional logic. CO2 (10)
- How Alpha-Beta pruning helps in overcoming the disadvantages of Minimax decisions. solve the following game tree using alpha beta method. Indicate the pruned branches very clearly in the diagram. CO2 (08)



- b) Explain the PEAS environment for WUMPUS world. CO2 (06)
 c) Explain how Artificial Intelligence can solve the following Tic Tac Toe problem. Discuss the concept with a game tree. CO2 (06)

X		O		X
O		X		
				O

UNIT - III

5. a) Explain the Unification used for reasoning under first order logic with example. CO3 (08)
 b) Differentiate forward chaining and backward chaining? CO3 (06)
 c) Describe the properties of planning graphs which are monotonically increasing or decreasing. CO3 (06)
6. a) Explain the forward chaining process in detail with example? What is the need of incremental chaining. CO3 (06)
 b) Write Forward (progression) state-space search algorithm. CO3 (06)
 c) Explain the process that converts any sentence into CNF. CO3 (08)

UNIT - IV

7. a) Given the full joint distribution as shown in fig below calculate the following: CO4 (08)
 i) $P(\text{toothache})$
 ii) $P(\text{Cavity})$
 iii) $P(\text{Toothache} \setminus \text{cavity})$
 iv) $P(\text{cavity} \setminus \text{toothache} \vee \text{catch})$

	Toothache		$\neg$ Toothache	
	Catch	$\neg$ Catch	Catch	$\neg$ Catch
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

- b) Explain how objects are organized in to categories, being the vital part of knowledge representation. CO4 (06)
 c) Justify how an agent forms rational decisions in an uncertain environment. CO4 (06)
8. a) With an example, explain categories and objects in knowledge representation. CO4 (10)
 b) Explain Event Calculus and the need for Event Calculus in Ontological Engineering CO4 (10)

UNIT - V

9. a) Illustrate how the squares known, frontier and other influences on handling uncertainty in wumpus world problem. CO5 (06)
 b) Discuss in brief the various robotic software architectures used in structuring algorithms. CO5 (08)
 c) Discuss any three application domains of robotic technology. CO5 (06)
10. a) Describe various kinds of sensors and its purpose in robots design. CO5 (10)
 b) Explain any five reason for uncertainty which effects the robot's actions and how uncertainty can be handled using robust control methods CO5 (10)
